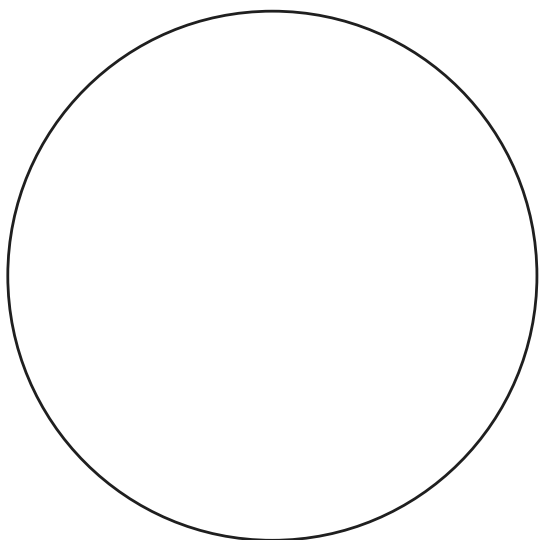
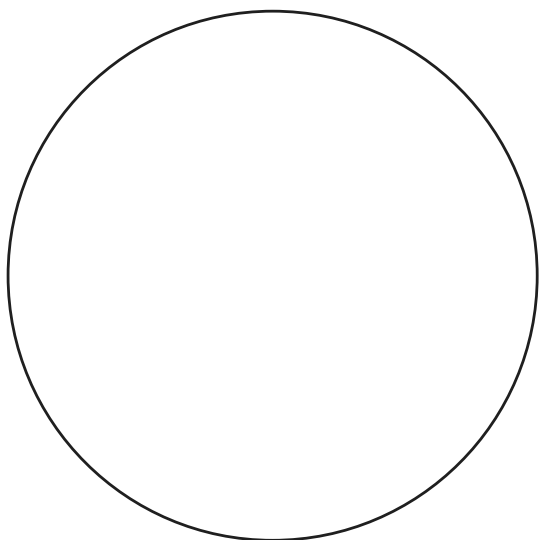
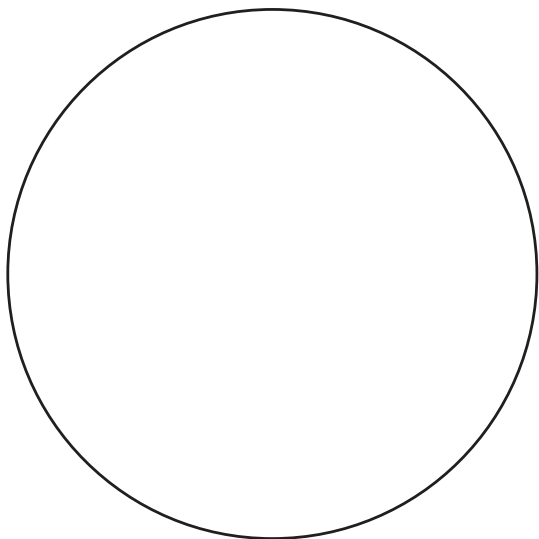
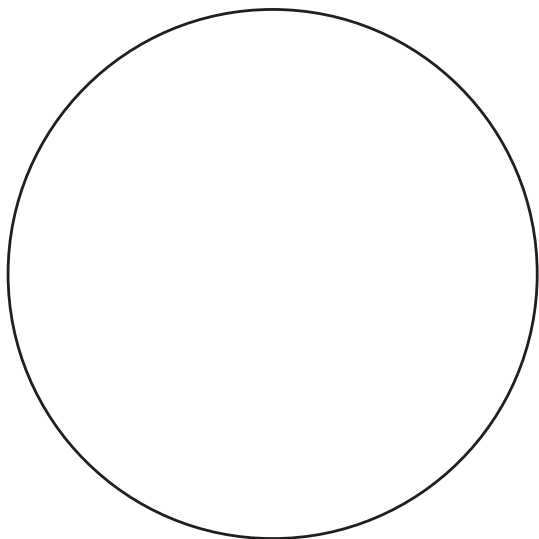
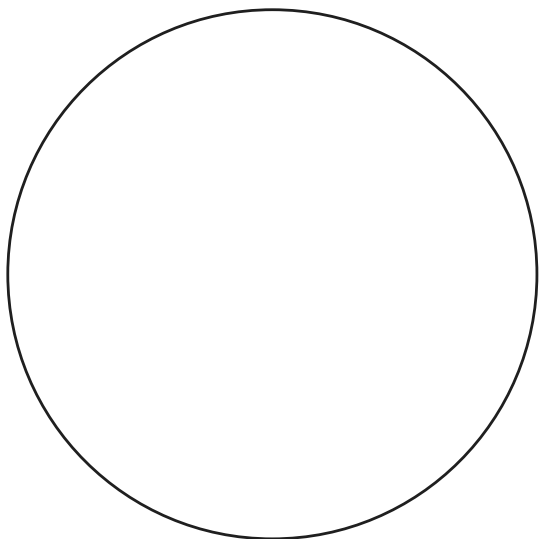
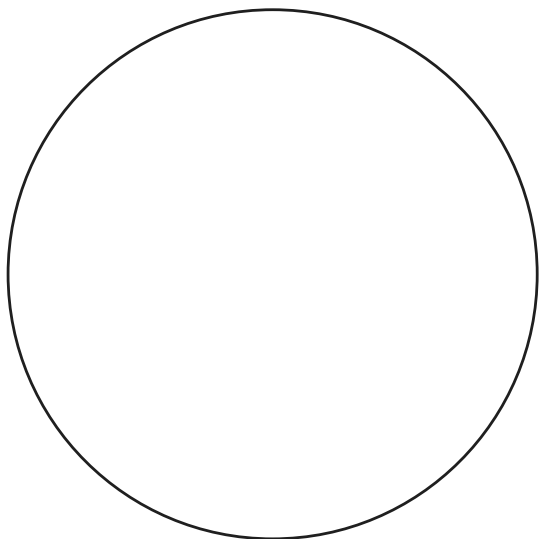
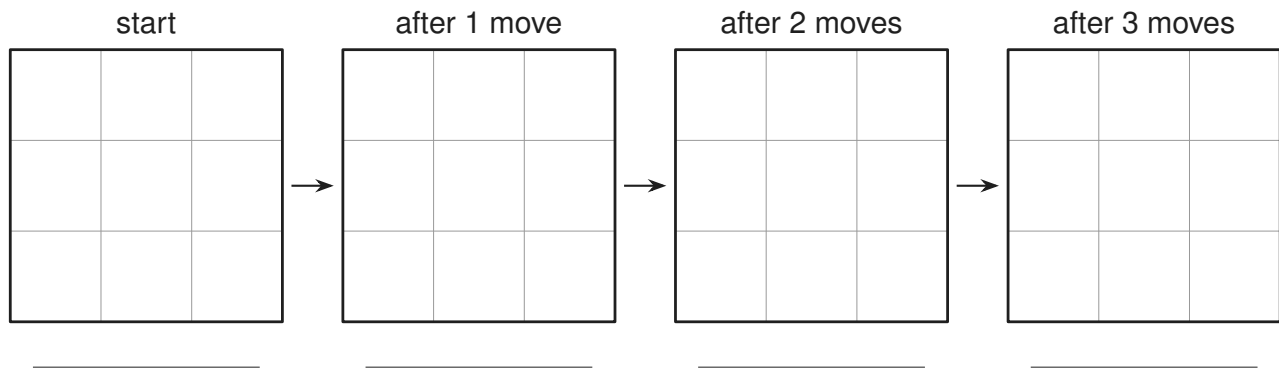
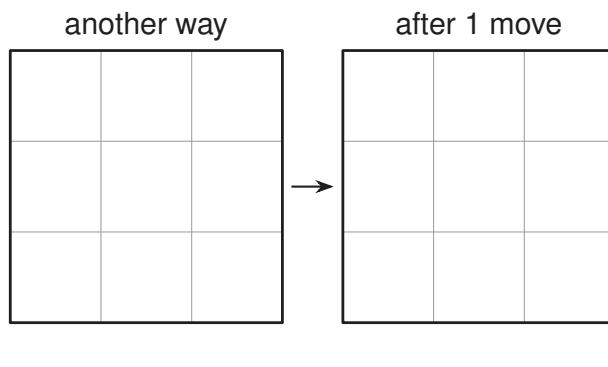
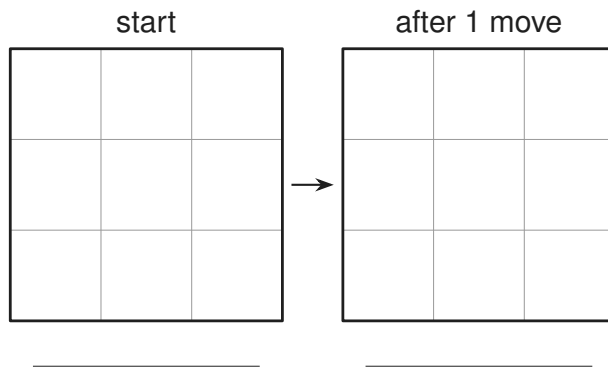
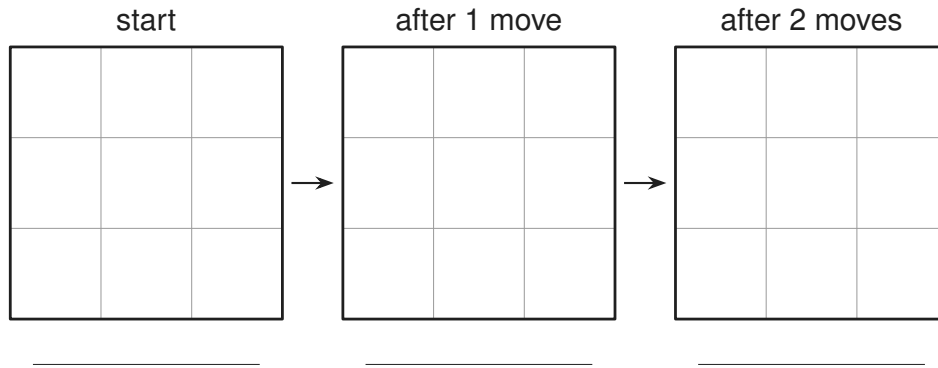




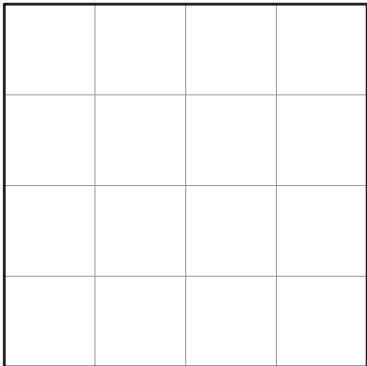
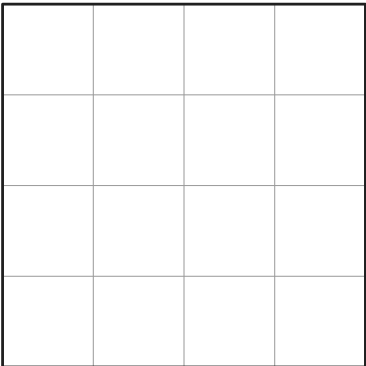
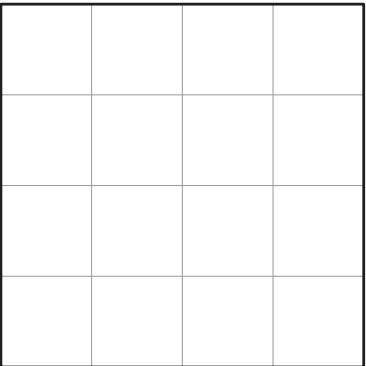
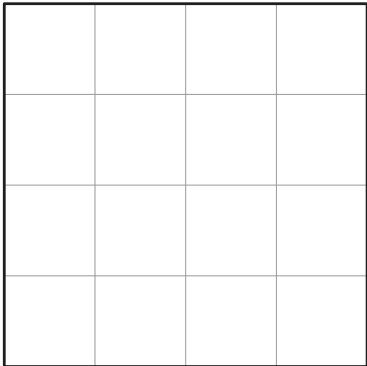
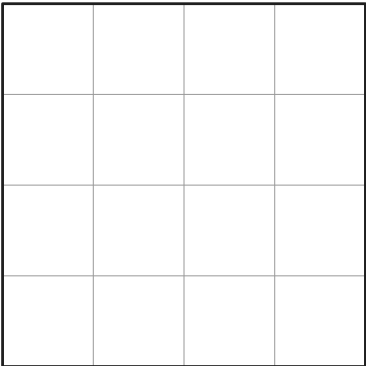
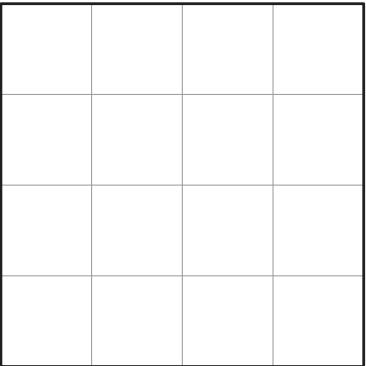
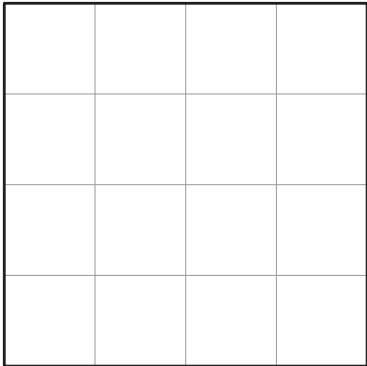
Problem 1: Put one counter on each of three plates. Color the start picture: for each plate, color one tower of squares, one square for each counter on that plate, starting at the bottom. Put the tallest tower on the left. Now make the move: take one counter off every plate that has counters, hold them all in one hand, and put them all together on one empty plate. Color the picture after 1 move. Make the move again and color the picture after 2 moves. Make the move a third time and color the picture after 3 moves. Count all the counters on the mat and say the number. Point to the first picture that looks just like the picture after it.



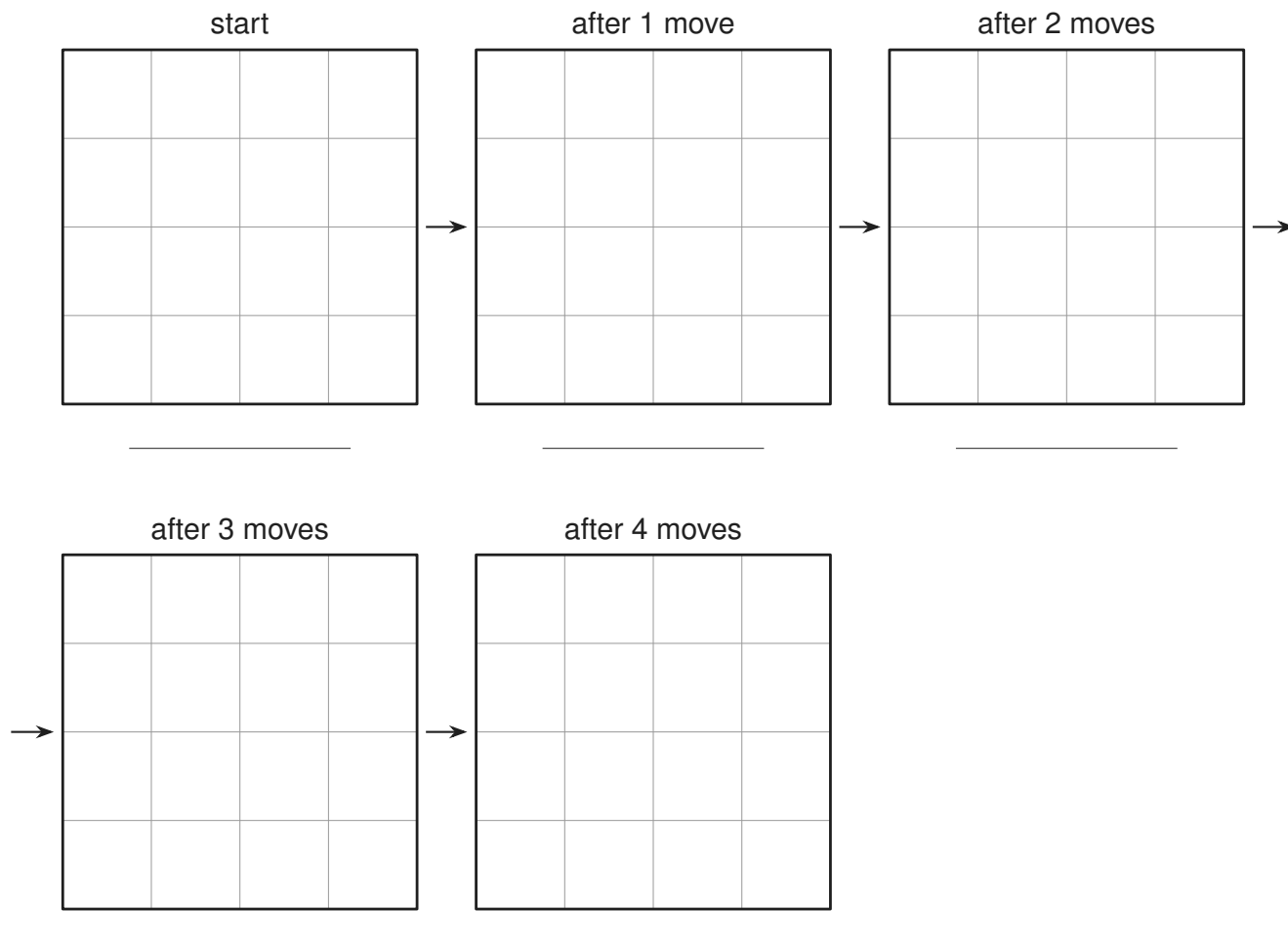
Problem 2: Put all three counters on one plate. Color the start picture. Make the move and color the picture after 1 move. Make the move again and color the picture after 2 moves. Now put two counters on one plate and one counter on another plate. Color the start picture. Make the move and color the picture after 1 move. Is there another way to put three counters on plates that you have not tried in Problems 1 and 2? If you find one, color it, make the move, and color what you get. Point to the picture where each start stopped changing.



Problem 3: Put four counters on one plate. Color the start picture. Make the move and color the picture after 1 move. Keep making moves and coloring a picture after each move until you have made six moves. Find two pictures that look the same and put a dot under both of them. Find more pairs. Does the picture ever stop changing?

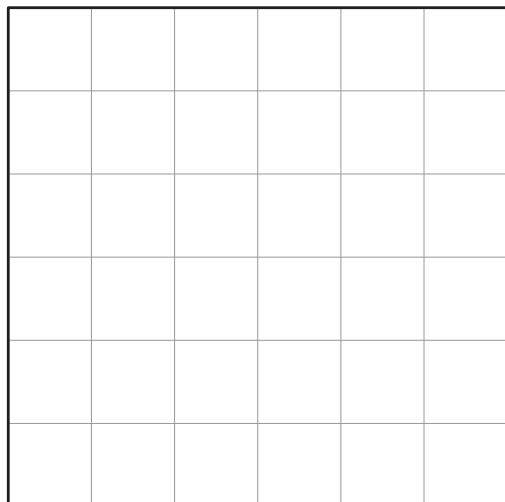
start	after 1 move	after 2 moves
		
_____	_____	_____
→		
after 3 moves	after 4 moves	after 5 moves
		
_____	_____	_____
→		
after 6 moves		
		

Problem 4: Put one counter on each of four plates. Color the start picture. Make four moves and color a picture after each one. Find the picture in Problem 3 that looks the same as your picture after 1 move, and point to it. Do both runs go round the same three pictures?

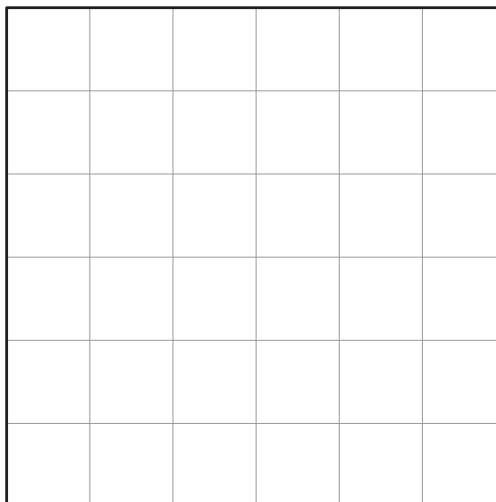


Problem 5: Put six counters on one plate. Color the start picture. Make the move and color the picture after 1 move. Keep making moves and coloring until a move changes nothing. Say how many moves you made before the picture stopped changing. Look at the last picture. Say how tall each tower is.

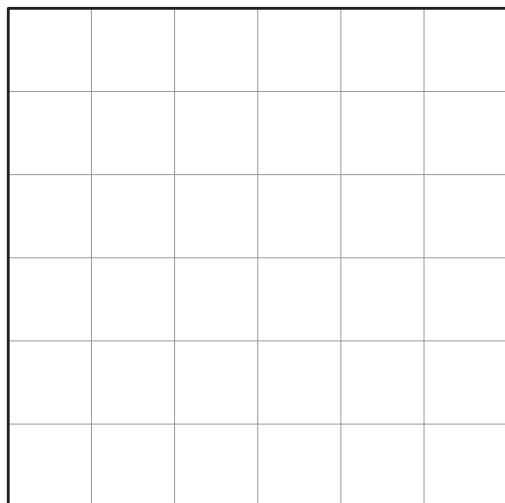
start



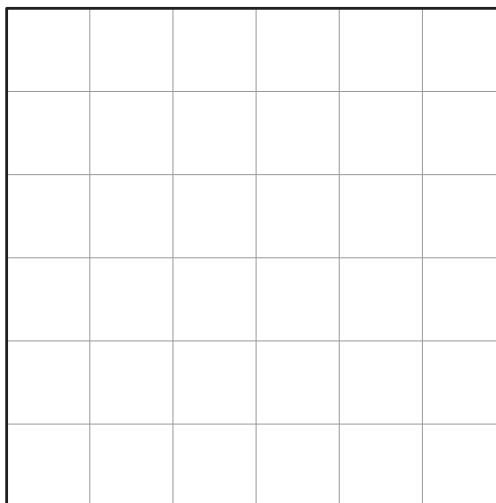
after 1 move



after 2 moves



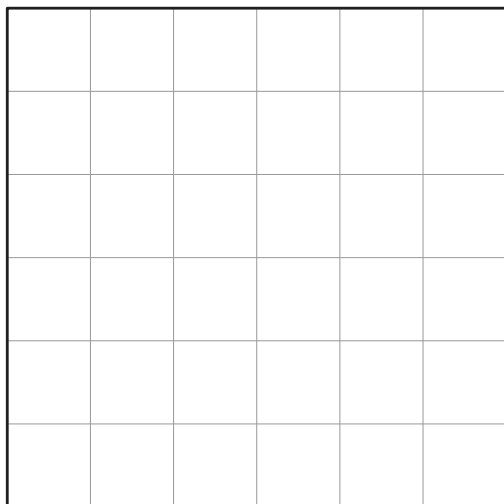
after 3 moves



moves: _____

Problem 6: Put one counter on each of six plates. Make one move and look at the plates. Which problem started like this? Keep making moves until a move changes nothing, and count the moves. Color the last picture. Is it the same as the last picture in Problem 5? Which run took more moves, this one or the one in Problem 5, and how many more?

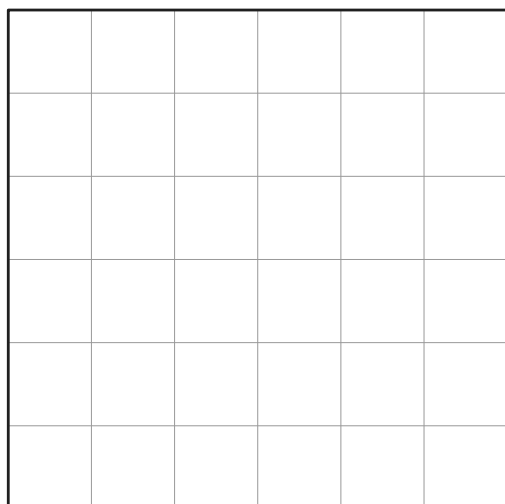
last picture



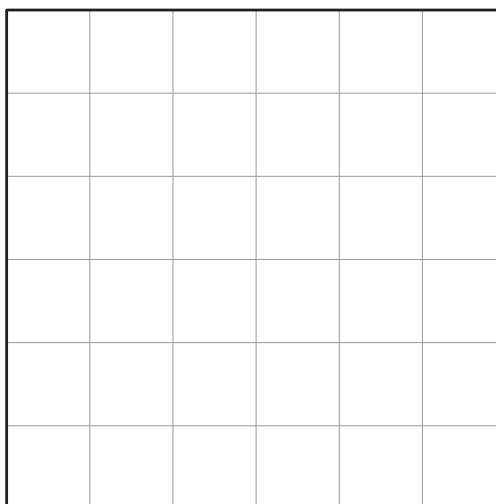
moves: _____

Problem 7: Put three counters on one plate and three counters on another plate. Color the start picture. Make moves and color a picture after each one until a move changes nothing. Count the moves. Now put two counters on each of three plates. Color the start picture. Make moves until a move changes nothing. Count the moves and color the last picture. Find the picture in the first run that looks the same as the start picture of the second run. Which run took more moves, and how many more?

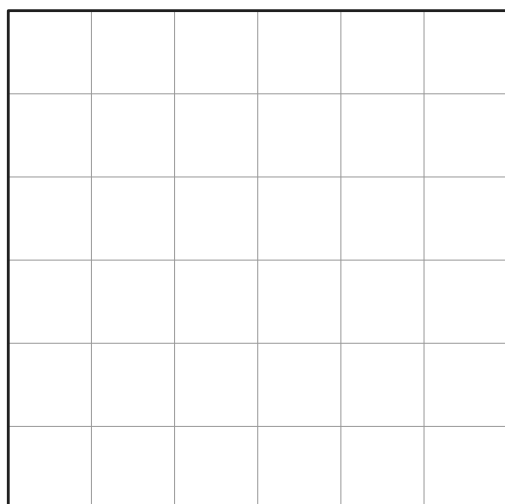
start



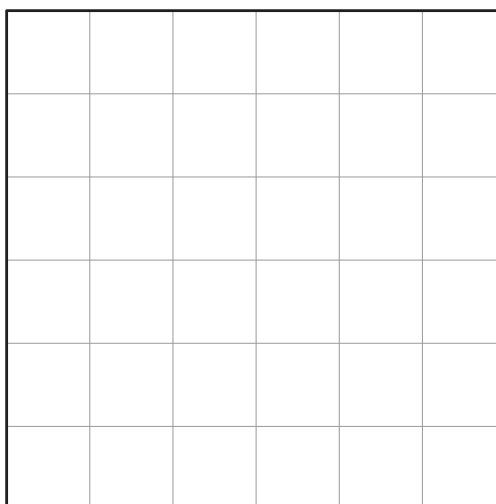
after 1 move



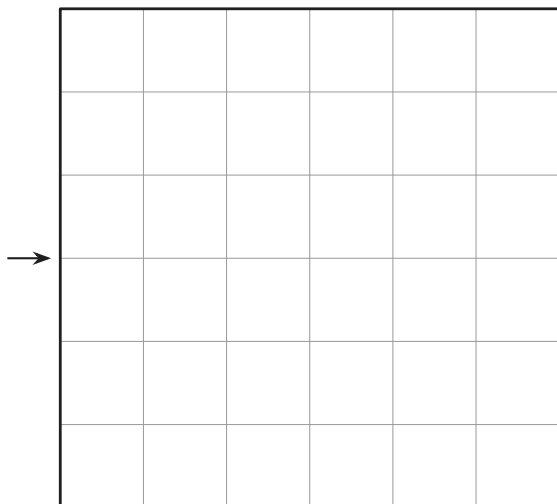
after 2 moves



after 3 moves

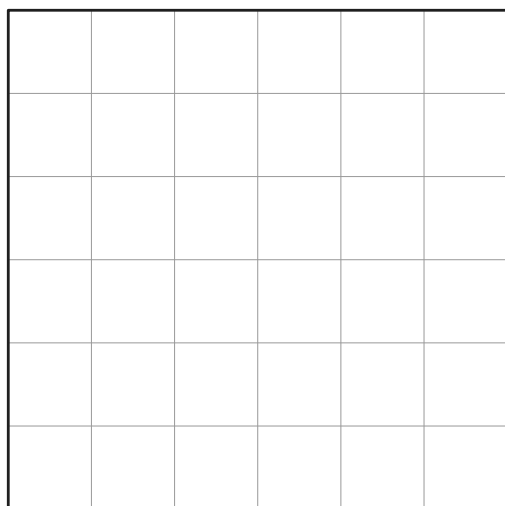


after 4 moves



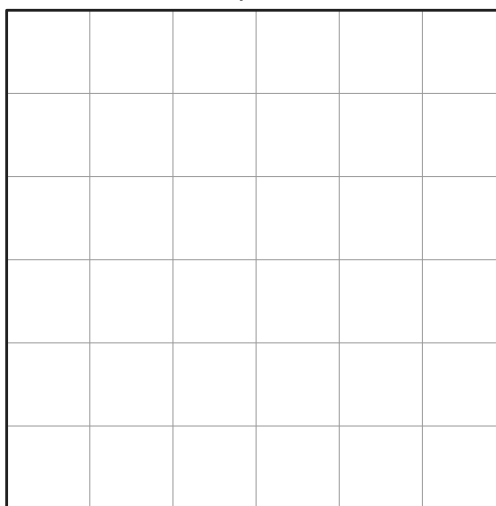
moves: _____

start



moves: _____

last picture

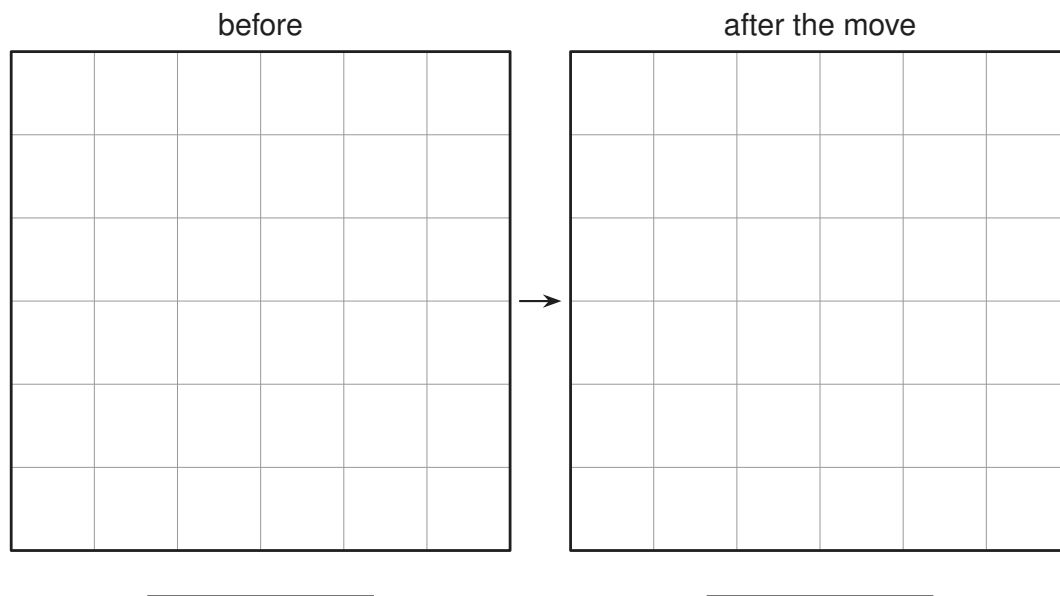
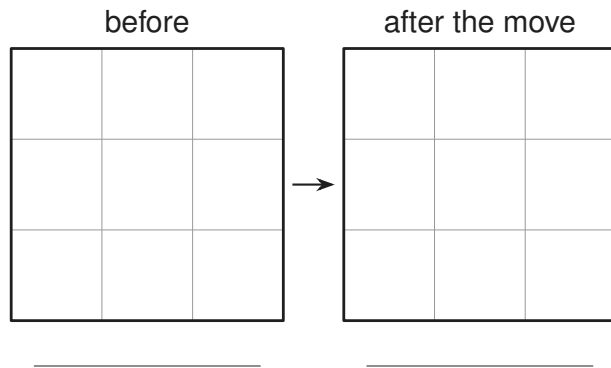


Problem 8: Put two counters on each of three plates. Before you make the move, count the plates that have counters on them and say the number. Make the move. Count the counters on the new plate and say the number. Do this before and after every move until a move changes nothing. Now put three counters on one plate and three on another. Before each move, say how many counters the new pile is going to have. Make the move and check. Keep going until a move changes nothing.

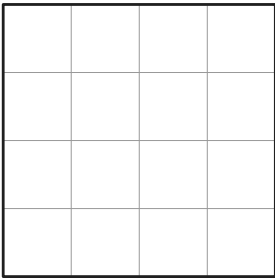
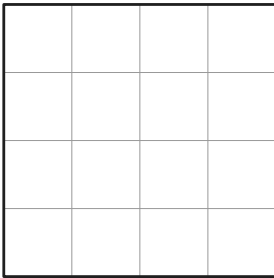
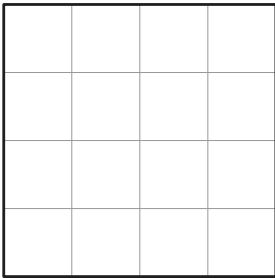
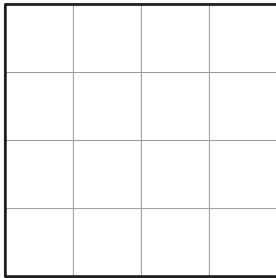
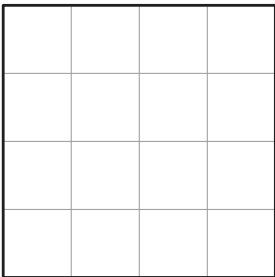
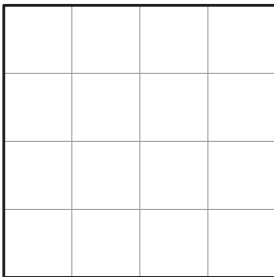
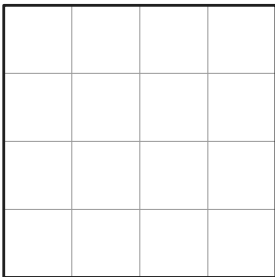
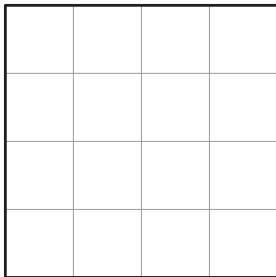
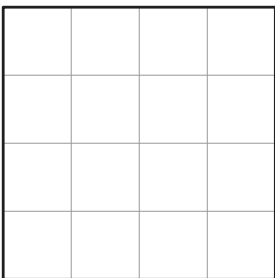
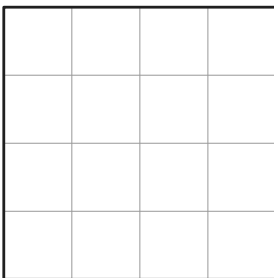
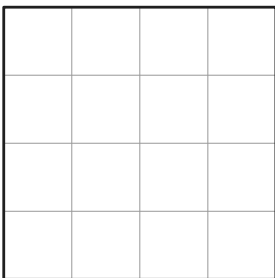
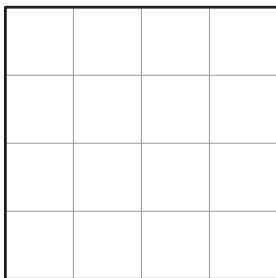
plates with counters	counters on the new plate

my guess for the new pile	the new pile

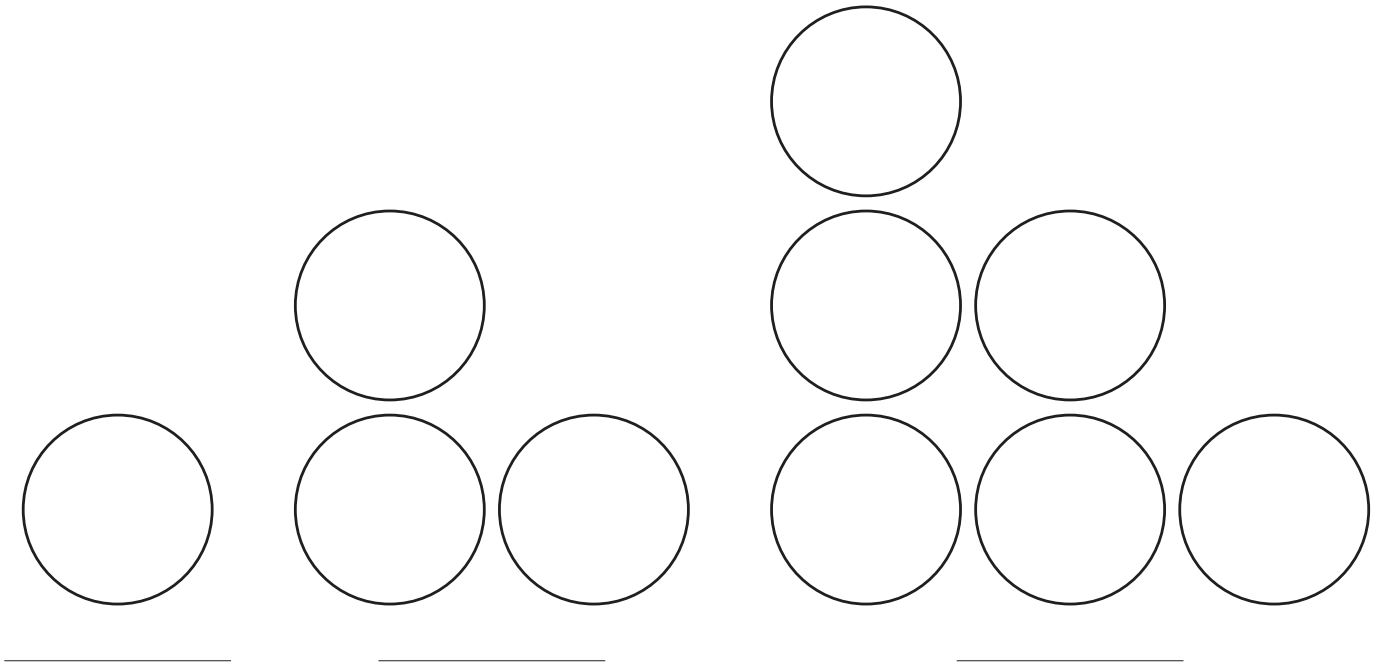
Problem 9: Put two counters on one plate and one counter on another plate. Color the picture. Make the move and color the picture after the move. Did the picture change? Now put three counters on one plate, two on a second plate and one on a third plate. Color the picture. Make the move and color the picture after the move. Did the picture change? Say what happened to the plate with three counters, to the plate with two, and to the plate with one, and say how many counters the new pile got and why it got that many.

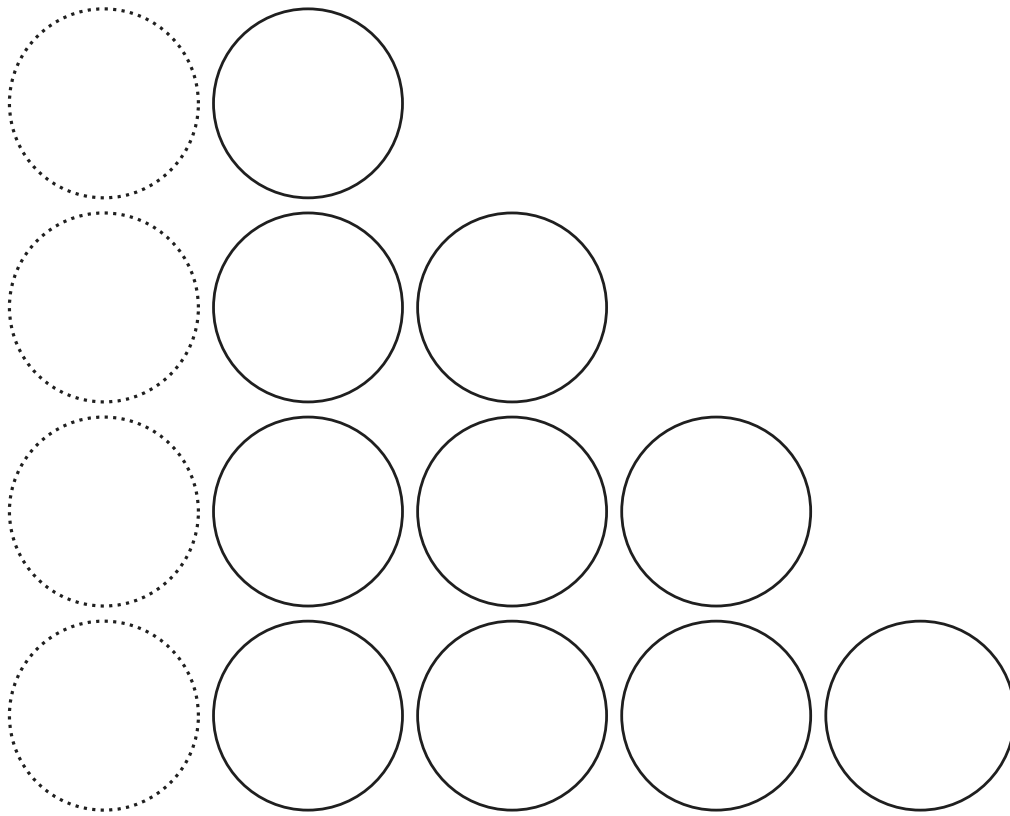


Problem 10: Find every way to put four counters on plates. Three counters on one plate and one on another plate is the same way as one and three. Color each way you find in its own frame on the left. When you think you have found them all, count the ways and say the number. Now build each way again, make one move, and color what you get in the frame on the right. Look at all the pictures. Did any way stay the same after the move? Which way on the left never appears on the right?

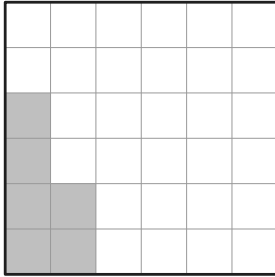
a way	after 1 move	a way	after 1 move
			
_____	_____	_____	_____
a way	after 1 move	a way	after 1 move
			
_____	_____	_____	_____
a way	after 1 move	a way	after 1 move
			
_____	_____	_____	_____

Problem 11: Cover every circle of the smallest staircase with counters. Count the counters and say the number. Do the same for the next staircase, then the next, then the biggest. Now leave the counters on the biggest staircase and make the move: take the top counter off every tower and build all of them into a new tower on the dotted circles. Do the towers look the same as before the move? Look at the four staircases. How many more counters does each staircase have than the one before it?

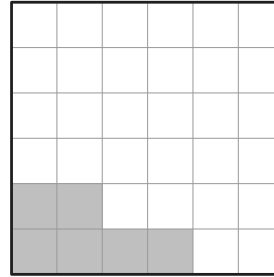




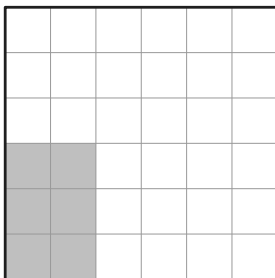
Problem 12: Everyone starts with six counters. First race: one of you starts with four counters on one plate and two on another; one starts with three and three; one starts with one counter on each of six plates. Make moves at the same time until a move changes nothing. Count your moves. Who needed the fewest moves? Second race: two, two, one and one; five and one; two, two and two. Count the moves. Now everyone: try to find a start that takes more moves than any start in the two races. Then find a start that takes only one move.



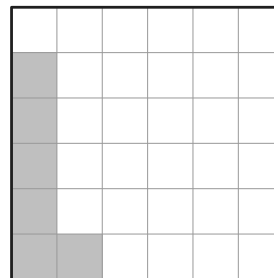
moves: _____



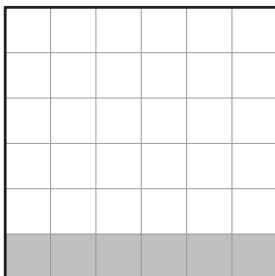
moves: _____



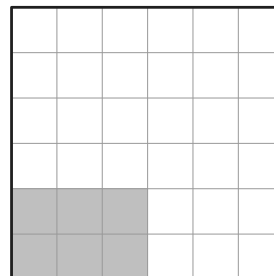
moves: _____



moves: _____

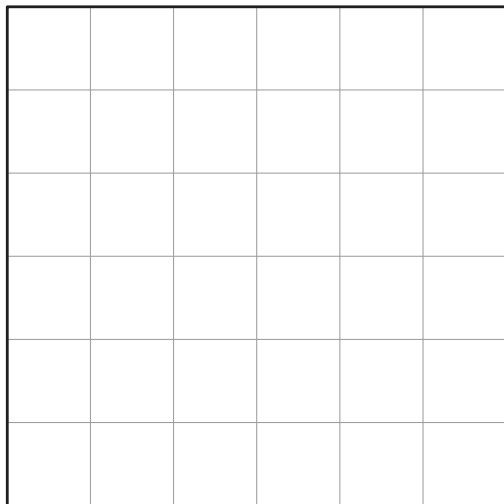


moves: _____



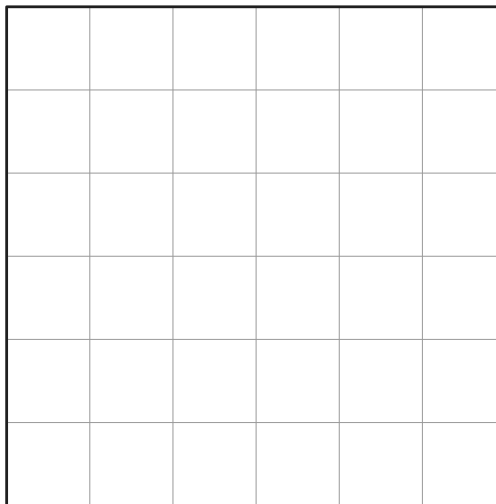
moves: _____

my start



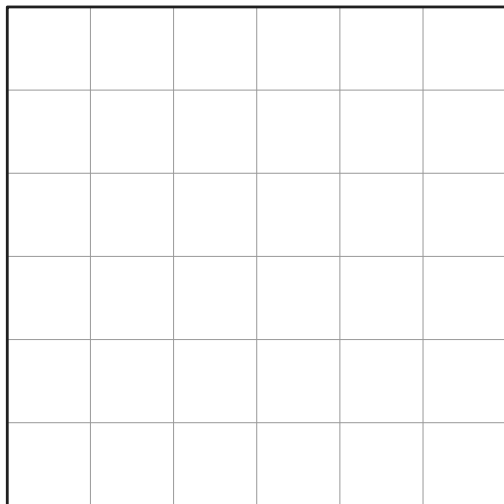
moves:

my start



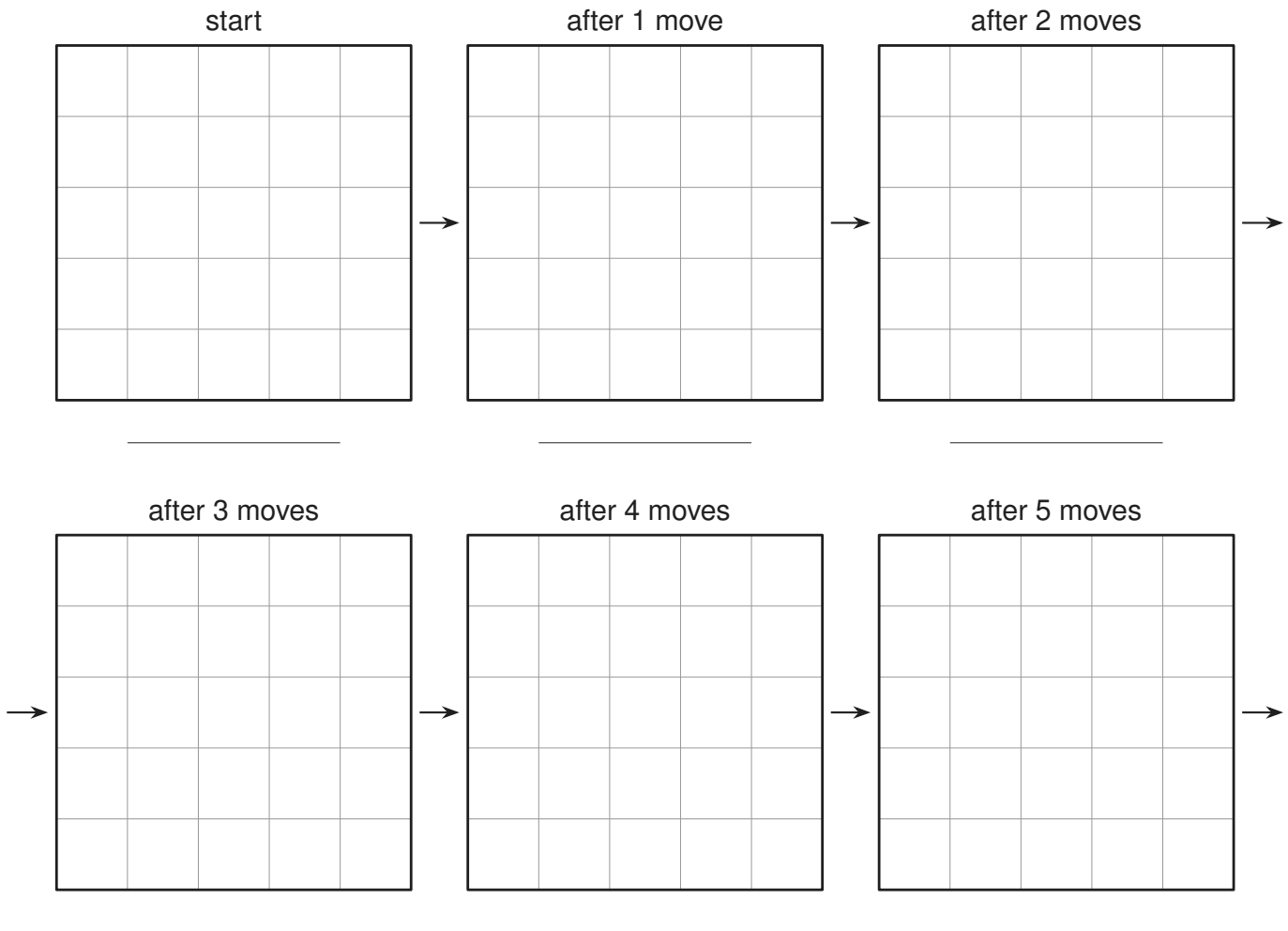
moves:

my start

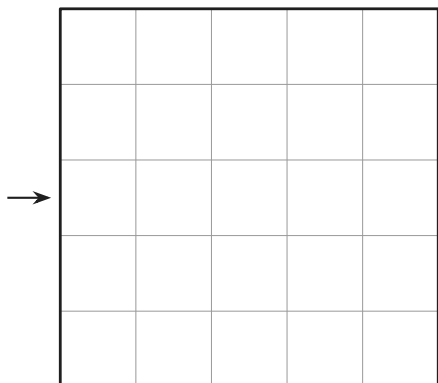


moves:

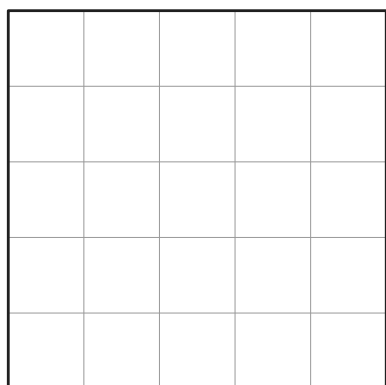
Problem 13: Put five counters on one plate. Color the start picture. Make moves and color a picture after each one until you have made six moves. Find pictures that are the same and put a dot under each of them. Then put one counter on each of five plates and make four moves, coloring a picture after each one. Point to the picture where this run joins the first run. Now think about three, four, five and six counters: which of them stop changing, and which go round and round?



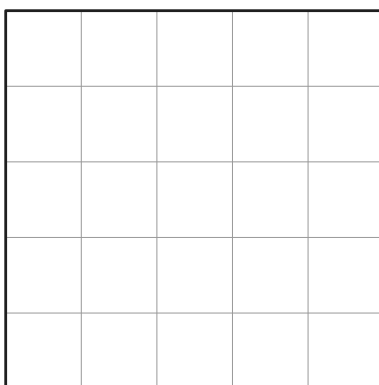
after 6 moves



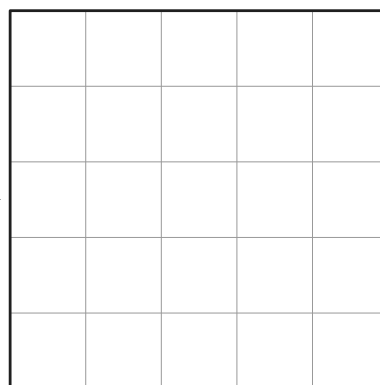
start



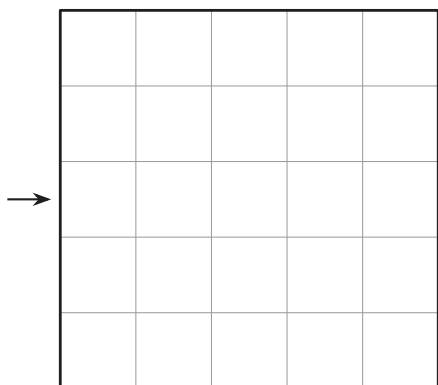
after 1 move



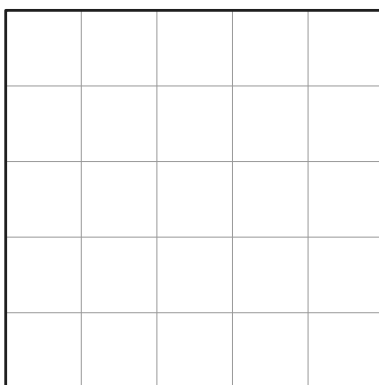
after 2 moves



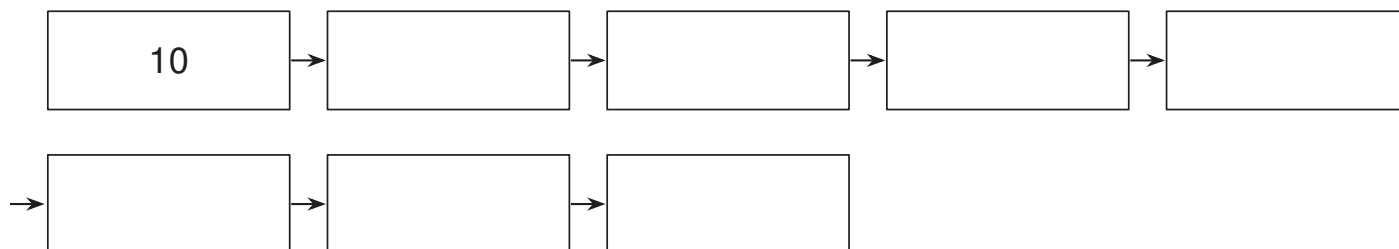
after 3 moves



after 4 moves

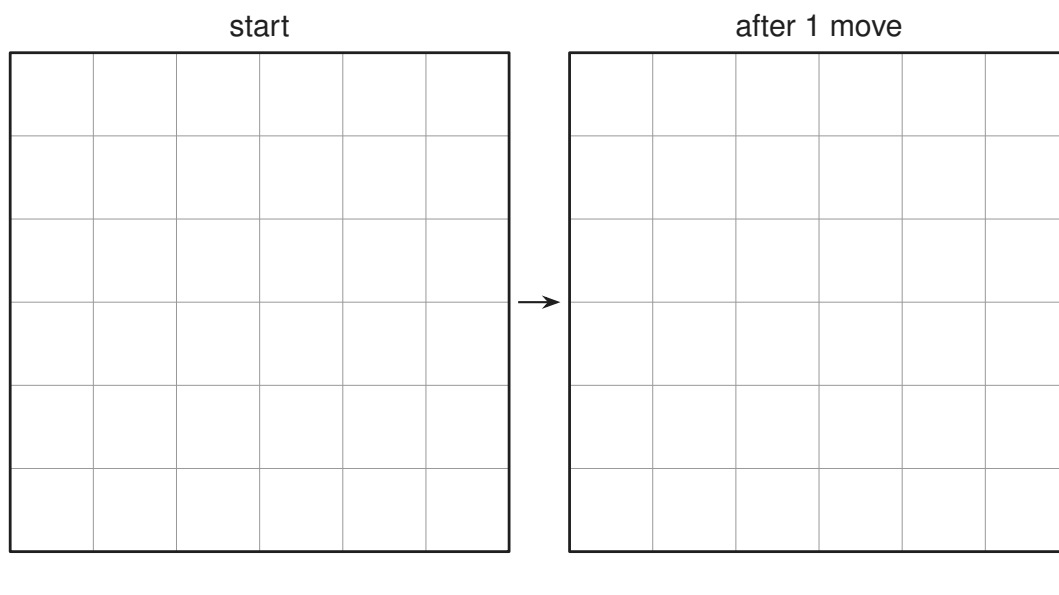
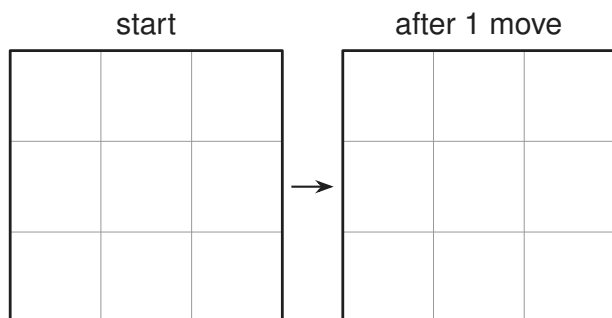


Problem 14: Put ten counters on one plate. Make the move. Say how tall each tower is, tallest first, so that it can be written in the next box. Keep making moves and saying the towers until a move changes nothing. How many moves did it take? Build the last picture on the big staircase board from Problem 11. Is it a staircase? One plate of three counters took 1 move to stop changing, and one plate of six took 3 moves. How many did one plate of ten take?

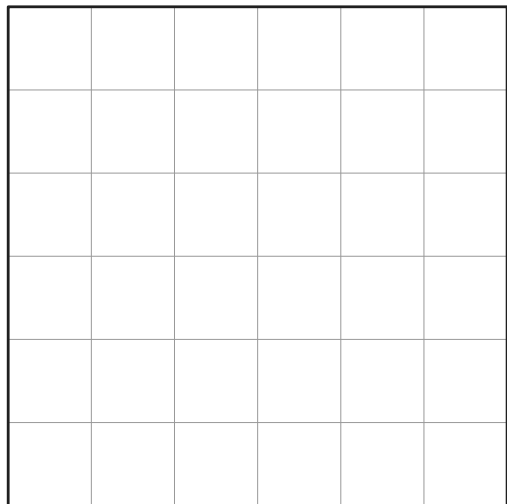


moves: _____

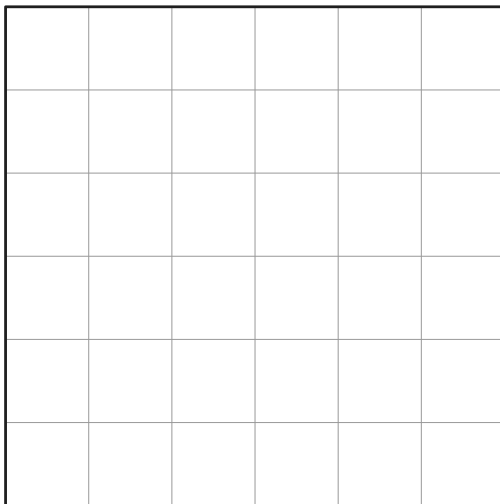
Problem 15: Find a start with three counters that becomes three counters on one plate after one move. Find a start with six counters that becomes five and one after one move. Find a different start that also becomes five and one. Find a start with six counters that becomes three, two and one after one move but is not three, two and one itself. Color each start you find on the left and what it became on the right.



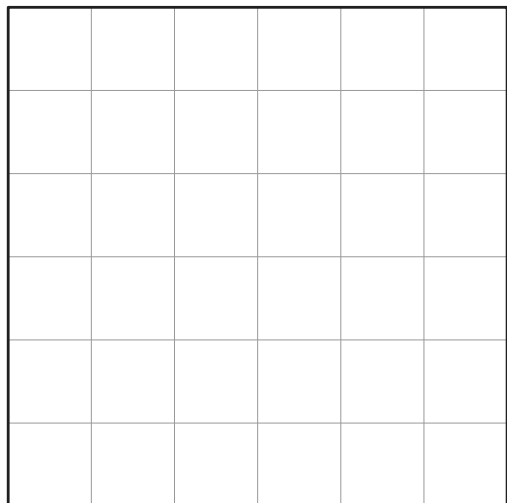
start



after 1 move



start



after 1 move

